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Photovoltaic Inverter and Power Adjustment Method Thereof, and Photovoltaic System

Abstract

A photovoltaic inverter includes a direct current conversion unit, an alternating current inverter unit, a control unit, and a direct current bus. An output end of the alternating current inverter unit is configured to connect to a load. The control unit is configured to: when a load power is less than a maximum power output by the photovoltaic module in a current illumination environment, control the direct current conversion unit to switch from a maximum power point tracking (MPPT) mode to a bus constant mode, so that an input voltage of the direct current conversion unit changes with the load power, and an output power of the photovoltaic module and an output power needed by the photovoltaic inverter keep being adjusted synchronously. The bus constant mode is to adjust an output voltage of the direct current conversion unit to a constant value or to be within a preset range.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a continuation of International Patent Application No. PCT/CN2023/131673 filed on Nov. 15, 2023, which claims priority to Chinese Patent Application No. 202211443749.4 filed on Nov. 18, 2022, both of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] This disclosure relates to the field of power electronics technologies, and in particular, to a photovoltaic inverter and a power adjustment method thereof, and a photovoltaic system.

BACKGROUND

[0003] A photovoltaic system is a new-type power generation system that converts solar radiation energy into electric energy for use by a mains power grid through photovoltaic effect of a photovoltaic solar panel. A photovoltaic inverter is a core part in the photovoltaic system, and is mainly responsible for converting a direct current (DC) generated by a photovoltaic module into an alternating current (AC), and outputting the alternating current to the mains power grid for use by the mains power grid. A two-stage inverter is a widely used photovoltaic inverter, and includes a direct current conversion unit and an alternating current inverter unit. The direct current conversion unit is mainly responsible for performing maximum power point tracking (MPPT) on the photovoltaic module, and the alternating current inverter unit is mainly responsible for connecting a maximum power tracked by the direct current conversion unit to the power grid. In a maximum power point tracking process, the direct current conversion unit usually perturbs a port voltage of the photovoltaic module at a fixed step size, to perform maximum power tracking. In a steady state, the direct current conversion unit runs near a maximum power point of the photovoltaic module.

[0004] A grid-connected power (that is, an input power of the power grid or an output power of the alternating current inverter unit) of the photovoltaic inverter has a requirement of rapid power change. In consideration of this, a requirement of rapid power adjustment is correspondingly raised for the direct current conversion unit. If the direct current conversion unit has a slow power control speed, an output power of the direct current conversion unit does not match the output power of the alternating current inverter unit. Currently, when the grid-connected power of the photovoltaic inverter may need to be lowered rapidly, the direct current conversion unit calculates a power variation, and controls the port voltage of the photovoltaic module at a specific step size to approach an open-circuit voltage of the photovoltaic module, to lower the output power of the direct current conversion unit. When the grid-connected power of the photovoltaic inverter may need to be increased rapidly, the direct current conversion unit calculates a power variation, and controls the port voltage of the photovoltaic module at a specific step size to approach a short-circuit voltage of the photovoltaic module, to increase the output power of the direct current conversion unit.

[0005] However, the direct current conversion unit controls the output power of the direct current conversion unit by sensing the power variation and then controlling the port voltage of the photovoltaic module at the specific step size. In this case, the following problems occur: A power

response speed is slow, and the output power of the direct current conversion unit cannot rapidly match the output power of the alternating current inverter unit.

SUMMARY

[0006] Embodiments of this disclosure provide a photovoltaic inverter and a power adjustment method thereof, and a photovoltaic system, to improve a power response speed of the photovoltaic inverter, so that an output power of a direct current conversion unit quickly matches an output power of an alternating current inverter unit.

[0007] According to a first aspect, this disclosure provides a photovoltaic inverter. The photovoltaic inverter includes a direct current conversion unit, an alternating current inverter unit, a control unit, and a direct current bus. An input end of the direct current conversion unit is configured to connect to a photovoltaic module. An output end of the direct current conversion unit is connected to an input end of the alternating current inverter unit through the direct current bus. An output end of the alternating current inverter unit is configured to connect to a load. The control unit is configured to: when a load power is less than a maximum power output by the photovoltaic module in a current illumination environment, control the direct current conversion unit to switch from an MPPT mode to a bus constant mode, so that an input voltage of the direct current conversion unit changes with the load power, and an output power of the photovoltaic module and an output power needed by the photovoltaic inverter keep being adjusted synchronously. The bus constant mode is to adjust an output voltage of the direct current conversion unit to a constant value or to be within a preset range. Therefore, when the load power is less than the maximum power output by the photovoltaic module in the current illumination environment, the control unit in this disclosure controls the direct current conversion unit to switch from the MPPT mode to the bus constant mode. In this way, the input voltage of the direct current conversion unit changes with the load power, to improve a power response speed of the direct current conversion unit, so that the output power of the direct current conversion unit can quickly match the output power of the alternating current inverter unit.

[0008] Optionally, any value within the preset range in this disclosure is greater than a peak value of an output voltage of the alternating current inverter unit. In this way, it can be ensured that the alternating current inverter unit can output a current. Any value within the preset range in this disclosure may also be less than a value. The value may be obtained based on experimental data in an actual application scenario.

[0009] With reference to the first aspect, in a possible implementation, when the direct current conversion unit is in the MPPT mode, the direct current conversion unit performs MPPT to maximize an output power of the direct current conversion unit. When performing MPPT, the direct current conversion unit may perturb a port voltage of the photovoltaic module (that is, the input voltage of the direct current conversion unit) at a fixed step size, to perform maximum power tracking. When the direct current conversion unit is in the bus constant mode, the direct current conversion unit adjusts the output voltage of the direct current conversion unit to the constant value or to be within the preset range, so that the bus voltage remains in a constant voltage state. The constant voltage state herein does not mean that the voltage is constant at a value. The constant voltage state herein is relative. Certainly, the constant voltage state may allow the voltage to fluctuate within a specific range. Therefore, the control unit is configured to: when controlling the direct current conversion unit to switch from the MPPT mode to the bus constant mode, switch from controlling the input voltage of the direct current conversion unit to controlling, by adjusting a duty cycle of a control signal of the direct current conversion unit, the output voltage of the direct current conversion unit to be adjusted to the constant value or to be within the preset range.

[0010] It may be understood that when the load power changes, the bus voltage fluctuates temporarily. For example, when the load power rises, a phenomenon in which the bus voltage drops transiently occurs. When the load power rises, a phenomenon in which the bus voltage rises transiently occurs.

[0011] With reference to the first aspect, in a possible implementation, when the load power is zero, the direct current conversion unit is in the bus constant mode, and the direct current conversion unit may increase the output voltage of the direct current conversion unit to a bus reference voltage value based on an input voltage provided by the photovoltaic module, to implement the bus constant mode. The bus reference voltage value is greater than a lower limit value of the preset range and less than an upper limit value of the preset range. In other words, the bus reference voltage value is a value within the preset range.

[0012] With reference to the first aspect, in a possible implementation, when the load power rises to be greater than zero and less than the maximum power output by the photovoltaic module in the current illumination environment, the direct current conversion unit is still in the bus constant mode. When detecting that the output voltage of the direct current conversion unit drops to be less than a bus reference voltage value, the direct current conversion unit increases the output power of the photovoltaic module, so that the output voltage of the direct current conversion unit rises to the bus reference voltage value, to implement the bus constant mode.

[0013] With reference to the first aspect, in a possible implementation, when the direct current conversion unit is in the bus constant mode and the load power drops, and the direct current conversion unit detects that the output voltage of the direct current conversion unit rises to be greater than the bus reference voltage value, the direct current conversion unit lowers the output power of the photovoltaic module, so that the output voltage of the direct current conversion unit drops to the bus reference voltage value, to implement the bus constant mode.

[0014] With reference to the first aspect, in a possible implementation, the control unit is further configured to: when the load power is greater than or equal to the maximum power output by the photovoltaic module in the current illumination environment, control the direct current conversion unit to switch from the bus constant mode to the MPPT mode to perform MPPT on the photovoltaic module.

[0015] Alternatively, when the direct current conversion unit is in the MPPT mode, and the alternating current inverter unit detects that the output voltage of the direct current conversion unit is not equal to the bus reference voltage value, the alternating current inverter unit lowers an output power of the alternating current inverter unit, to adjust the output voltage of the direct current conversion unit to the bus reference voltage value.

[0016] According to a second aspect, this disclosure provides a power adjustment method for a photovoltaic inverter. The method is applied to the photovoltaic inverter. The photovoltaic inverter includes a direct current conversion unit, an alternating current inverter unit, and a direct current bus. An input end of the direct current conversion unit is configured to connect to a photovoltaic module. An output end of the direct current conversion unit is connected to an input end of the alternating current inverter unit through the direct current bus. An output end of the alternating current inverter unit is configured to connect to a load. This method includes: detecting a load power and a maximum power output by the photovoltaic module in a current illumination environment; and when the load power is less than the maximum power output by the photovoltaic module in the current illumination environment, controlling the direct current conversion unit to switch from an MPPT mode to a bus constant mode, so that an input voltage of the direct current conversion unit changes with the load power, and an output power of the photovoltaic module and an output power needed by the photovoltaic inverter keep being adjusted synchronously. The bus constant mode is to adjust an output voltage of the direct current conversion unit to a constant value or to be within a preset range.

[0017] In this disclosure, the load power and the maximum power output by the photovoltaic module in the current illumination environment are monitored in real time. Further, when the load power is less than the maximum power output by the photovoltaic module in the current illumination environment, the direct current conversion unit is controlled to switch from the MPPT mode to the bus constant mode, so that the input voltage of the direct current conversion unit

changes with the load power, and the output power of the photovoltaic module is consistent with the output power needed by the photovoltaic inverter. In this way, the power response speed of the photovoltaic inverter is improved, so that the output power of the direct current conversion unit can quickly match the load power.

[0018] Optionally, any value within the preset range in this disclosure is greater than a peak value of an output voltage of the alternating current inverter unit. In this way, it can be ensured that the alternating current inverter unit can output a current. Any value within the preset range in this disclosure may also be less than a value. The value may be obtained based on experimental data in an actual application scenario.

[0019] It should be noted that when the direct current conversion unit is in the MPPT mode, the direct current conversion unit performs MPPT to maximize an output power of the direct current conversion unit. When performing MPPT, the direct current conversion unit may perturb a port voltage of the photovoltaic module (that is, the input voltage of the direct current conversion unit) at a fixed step size, to perform maximum power tracking. When the direct current conversion unit is in the bus constant mode, the direct current conversion unit adjusts the output voltage of the direct current conversion unit to the constant value or to be within the preset range, so that the bus voltage remains in a constant voltage state. The constant voltage state herein does not mean that the voltage is constant at a value. The constant voltage state herein is relative. Certainly, the constant voltage state may allow the voltage to fluctuate within a specific range.

[0020] With reference to the second aspect, in a possible implementation, the controlling the direct current conversion unit to switch from an MPPT mode to a bus constant mode includes: when controlling the direct current conversion unit to switch from the MPPT mode to the bus constant mode, adjusting a duty cycle of a control signal of the direct current conversion unit, to switch from controlling the input voltage of the direct current conversion unit to controlling the output voltage of the direct current conversion unit to be adjusted to the constant value or to be within the preset range.

[0021] It may be understood that when the load power changes, the bus voltage fluctuates temporarily. For example, when the load power rises, a phenomenon in which the bus voltage drops transiently occurs. When the load power rises, a phenomenon in which the bus voltage rises transiently occurs.

[0022] With reference to the second aspect, in a possible implementation, after the controlling the direct current conversion unit to switch from an MPPT mode to a bus constant mode, the method further includes: when the load power is zero, increasing the output voltage of the direct current conversion unit to a bus reference voltage value based on an input voltage provided by the photovoltaic module, to implement the bus constant mode. The bus reference voltage value is greater than a lower limit value of the preset range and less than an upper limit value of the preset range. In other words, the bus reference voltage value is a value within the preset range.

[0023] With reference to the second aspect, in a possible implementation, after the controlling the direct current conversion unit to switch from an MPPT mode to a bus constant mode, the method further includes: when the load power rises to be greater than zero and less than the maximum power output by the photovoltaic module in the current illumination environment, and it is detected that the output voltage of the direct current conversion unit drops to be less than a bus reference voltage value, increasing the output power of the photovoltaic module, so that the output voltage of the direct current conversion unit rises to the bus reference voltage value, to implement the bus constant mode.

[0024] With reference to the second aspect, in a possible implementation, after the controlling the direct current conversion unit to switch from an MPPT mode to a bus constant mode, the method further includes: when the load power drops, and it is detected that the output voltage of the direct current conversion unit rises to be greater than the bus reference voltage value, lowering the output power of the photovoltaic module, so that the output voltage of the direct current conversion unit

drops to the bus reference voltage value, to implement the bus constant mode.

[0025] With reference to the second aspect, in a possible implementation, the method further includes: when the load power is greater than or equal to the maximum power output by the photovoltaic module in the current illumination environment, controlling the direct current conversion unit to switch from the bus constant mode to the MPPT mode to perform MPPT on the photovoltaic module.

[0026] Alternatively, after the controlling the direct current conversion unit to switch from the bus constant mode to the MPPT mode, the method further includes: when it is detected that the output voltage of the direct current conversion unit is not equal to the bus reference voltage value, lowering an output power of the alternating current inverter unit, to adjust the output voltage of the direct current conversion unit to the bus reference voltage value.

[0027] Alternatively, after the lowering an output power of the alternating current inverter unit, to adjust the output voltage of the direct current conversion unit to the bus reference voltage value, the method further includes: controlling a port voltage of the photovoltaic module to drop to a target voltage value, where the target voltage value is less than a voltage value corresponding to the maximum power output by the photovoltaic module in the current illumination environment. In this way, the output power of the photovoltaic module can be lowered. In addition, a possibility that the direct current conversion unit and the alternating current inverter unit fail can be reduced.

[0028] According to a third aspect, this disclosure provides a photovoltaic system. The photovoltaic system includes a photovoltaic module, and the photovoltaic inverter according to the first aspect or any one of the possible implementations of the first aspect, or the photovoltaic inverter that uses the method according to the second aspect or any one of the possible implementations of the second aspect. An input end of the photovoltaic inverter is connected to the photovoltaic module. An output end of the photovoltaic inverter is configured to connect to a load. The photovoltaic inverter is configured to: convert a direct current of the photovoltaic module into an alternating current and input the alternating current to the load; and when a load power is less than a maximum power output by the photovoltaic module in a current illumination environment, control the direct current conversion unit to switch from an MPPT mode to a bus constant mode, so that an input voltage of the direct current conversion unit changes with the load power, and an output power of the photovoltaic module and an output power needed by the photovoltaic inverter keep being adjusted synchronously. The bus constant mode is to adjust an output voltage of the direct current conversion unit to a constant value or to be within a preset range.

[0029] It may be understood that implementations and technical effects of the foregoing aspects may be mutually referenced. Details are not described herein again.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0030] FIG. 1 is a diagram of an application scenario of a photovoltaic system according to an embodiment of this disclosure;

[0031] FIG. 2 is a diagram of a structure of a photovoltaic system according to an embodiment of this disclosure;

[0032] FIG. 3 is a diagram of another structure of a photovoltaic system according to an embodiment of this disclosure;

[0033] FIG. 4 is a diagram of still another structure of a photovoltaic system according to an embodiment of this disclosure;

[0034] FIG. 5 is a diagram of a P-U curve of a photovoltaic module according to an embodiment of this disclosure;

[0035] FIG. 6 is a diagram of a bus voltage-time curve according to an embodiment of this

disclosure; and

[0036] FIG. 7 is a schematic flowchart of a power adjustment method for a photovoltaic inverter according to an embodiment of this disclosure.

DESCRIPTION OF EMBODIMENTS

[0037] In the descriptions of this disclosure, unless otherwise specified, “/” generally indicates an “or” relationship between associated objects. For example, A/B may indicate A or B. A term “and/or” in this specification describes only an association relationship between associated objects and indicates that there may be three relationships. For example, A and/or B may represent the following three cases: Only A exist.sub.5, both A and B exist, and only B exists.

[0038] Solar energy is a type of inexhaustible and pollution-free green energy from nature. In other words, the solar energy is a type of renewable and clean energy that is widely used in lives and work of people. One way of using the solar energy is to convert the solar energy into electric energy. Solar power generation can be divided into solar thermal power generation and photovoltaic power generation. Photovoltaic power generation (photovoltaic power generation) is a technology of converting light energy into electric energy through photovoltaic effect of a semiconductor interface. A photovoltaic system provided in this disclosure may be a photovoltaic system based on solar photovoltaic power generation (for ease of description, referred to as a photovoltaic system below). Solar photovoltaic power generation features absence of a moving part, noise, and pollution, and high reliability, and has extensive application prospect in a communication photovoltaic system in a remote region.

[0039] The photovoltaic system provided in this disclosure is applicable to power supply of a plurality of types of power-consuming devices such as a base station device in a remote region with no or poor mains electricity, a battery, or a household device (for example, a refrigerator or an air conditioner). Specific application may be determined based on an actual application scenario. This is not limited herein. A photovoltaic inverter provided in this disclosure is applicable to the photovoltaic system; may be configured to convert a variable direct current generated by a direct current power supply such as a photovoltaic module into an alternating current of a mains electricity frequency, and output the alternating current to a mains power grid for use by the mains power grid; and specifically, may be used by a base station device, a battery, or a household device in the mains power grid. The photovoltaic system provided in this disclosure may be applied to different application scenarios, for example, a photovoltaic power supply scenario and a hybrid scenario of photovoltaic power supply and energy storage battery power supply. A specific scenario may be determined based on an actual application scenario. This is not limited herein. In the photovoltaic power supply scenario, a power supply for supplying power is a photovoltaic module. In the photovoltaic-and-storage hybrid power supply scenario, a power supply for supplying power includes a photovoltaic module and an energy storage battery string. This disclosure is described by using the photovoltaic power supply scenario as an example.

[0040] FIG. 1 is a diagram of an application scenario of a photovoltaic system according to an embodiment of this disclosure. In a photovoltaic power supply scenario, the photovoltaic system shown in FIG. 1 may include a photovoltaic module and a photovoltaic inverter. An output end of the photovoltaic module is connected to an input end of the photovoltaic inverter. An output end of the photovoltaic inverter is connected to a load such as an alternating current power grid, a household device, or a communication base station. Optionally, there may be a plurality of photovoltaic modules connected to the input end of the photovoltaic inverter herein. The plurality of photovoltaic modules may be connected in series and/or in parallel and then connected to the photovoltaic inverter. The photovoltaic module herein may be a solar panel, a photovoltaic panel, or the like. After the photovoltaic system starts to run, the photovoltaic inverter may convert a direct current input by the photovoltaic module into an alternating current and transmit the alternating current to a load for use by the load such as the alternating current power grid, the communication base station, or the household device.

[0041] The foregoing is merely an example of the application scenario of the photovoltaic system provided in this disclosure, but is not exhaustive. The application scenario is not limited in this disclosure.

[0042] With reference to FIG. 2 to FIG. 7, the following describes, by using examples, a photovoltaic system, a photovoltaic inverter, and a power adjustment method for the photovoltaic inverter provided in this disclosure.

[0043] For ease of description, the following is described by using an example in which a load connected to an output end of the photovoltaic inverter is an alternating current power grid.

[0044] FIG. 2 is a diagram of a structure of a photovoltaic system according to an embodiment of this disclosure. As shown in FIG. 2, the photovoltaic system **1** may include a photovoltaic module **10** and a photovoltaic inverter **20**. The photovoltaic module **10** is connected to an input end of the photovoltaic inverter **20**, and is configured to provide electric energy (that is, to supply power) for the photovoltaic inverter **20**, for example, provide an input current, an input voltage, or an input power. An output end of the photovoltaic inverter **20** may be connected to a load (for example, an alternating current power grid or a household device). The photovoltaic inverter is configured to convert a direct current input by the photovoltaic module **10** into an alternating current and transmit the alternating current to the load for use by the load.

[0045] In the photovoltaic system **1** shown in FIG. 2, the photovoltaic inverter **20** may include a direct current conversion unit **201**, an alternating current inverter unit **202**, a control unit **203**, and a direct current bus **204**. An input end of the direct current conversion unit **201** is connected to the photovoltaic module **10**. An output end of the direct current conversion unit **201** may be connected to an input end of the alternating current inverter unit **202** through the direct current bus **204**. An output end of the alternating current inverter unit **202** is connected to the load. The control unit **203** may be connected to the output end of the alternating current inverter unit **202**, and can monitor an output power (that is, an input power of the load, which is referred to as a load power in this disclosure, where details are not described below again) of the alternating current inverter unit **202** in real time. The control unit **203** may be further connected to the direct current conversion unit **201**, and can obtain, from the direct current conversion unit **201**, a maximum power output by the photovoltaic module **10** in a current illumination environment and control the direct current conversion unit **201** to perform mode switching.

[0046] It may be understood that, in the photovoltaic system **1** shown in FIG. 2, $U_{sub.pv}$ represents an output voltage of the photovoltaic module **10**, $U_{sub.bus}$ represents a bus voltage on the direct current bus **204**, and $U_{sub.ac}$ represents an output voltage of the alternating current inverter unit **202**.

[0047] Optionally, the direct current bus **204** may include one bus capacitor or a plurality of bus capacitors connected in series to each other, and may be configured to store energy. FIG. 3 is a diagram of another structure of a photovoltaic system according to an embodiment of this disclosure. In the photovoltaic system **1** shown in FIG. 3, the direct current bus **204** includes a bus capacitor C. The direct current conversion unit **201** may perform processing such as rectification or voltage step-up on a direct current output by the photovoltaic module **10**, and output a processed direct current and store the processed direct current on two ends of the bus capacitor C. The alternating current inverter unit **202** may convert the direct current on the two ends of the bus capacitor C into an alternating current of a corresponding power for use by the load.

[0048] In some feasible implementations, as shown in FIG. 3, the direct current conversion unit **201** may include a DC/DC conversion circuit **2011** and a DC/DC control circuit **2012**. The alternating current inverter unit **202** may include DC/AC conversion circuit **2021** and a DC/AC control circuit **2022**. An input end of the DC/DC conversion circuit **2011** is connected to the photovoltaic module **10**. An output end of the DC/DC conversion circuit **2011** is connected to an input end of the DC/AC conversion circuit **2021** through the direct current bus **204**. An output end of the DC/AC conversion circuit **2021** is connected to the load.

[0049] The DC/DC control circuit **2012** and the DC/AC control circuit **2022** may be connected, to exchange information. The DC/DC control circuit **2012** may be configured to connect to the photovoltaic module **10**, and can monitor and/or control one or more of the output voltage (U.sub.pv), an output current, and an output power of the photovoltaic module **10** in real time. The DC/DC control circuit **2012** may be further connected to the bus capacitor C on the direct current bus **204**, and can monitor a voltage of the bus capacitor C in real time, that is, the bus voltage (U.sub.bus). The DC/DC control circuit **2012** may be further connected to the DC/DC conversion circuit **2011**, and can control an output voltage of the DC/DC conversion circuit **2011**. The DC/AC control circuit **2022** may also be connected to the bus capacitor C on the direct current bus **204**, and can monitor the voltage of the bus capacitor C in real time, that is, the bus voltage (U.sub.bus). The DC/AC control circuit **2022** may be further connected to the DC/AC conversion circuit **2021**, and can monitor and control an output power of the DC/AC conversion circuit **2021** (that is, the load power) in real time.

[0050] Optionally, in the photovoltaic system **1** shown in each of FIG. 2 and FIG. 3, the direct current conversion unit **201** of the photovoltaic inverter **20** may be a DC/DC converter that can perform processing such as rectification or voltage step-up on a direct current input by the photovoltaic module **10** and then output a processed direct current. Optionally, the alternating current inverter unit **202** of the photovoltaic inverter **20** may be a DC/AC converter that can convert, into an alternating current, a direct current output by the DC/DC converter. The control unit **203** of the photovoltaic inverter **20** may be a controller, a control chip, a control system, an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or another programmable logic device, a discrete gate, a transistor logic device, or the like.

[0051] FIG. 4 is a diagram of still another structure of a photovoltaic system according to an embodiment of this disclosure. In the photovoltaic system **1** shown in FIG. 4, the photovoltaic inverter **20** includes a DC/DC converter **201**, a DC/AC converter **202**, a direct current bus **204**, and a controller **203**. An input end of the DC/DC converter **201** is connected to the photovoltaic module **10**. An output end of the DC/DC converter **201** may be connected to an input end of the DC/AC converter **202** through the direct current bus **204**. An output end of the DC/AC converter **202** is connected to an alternating current power grid. The DC/DC converter **201** may perform processing such as rectification or voltage step-up on a direct current output by the photovoltaic module **10**, to obtain a direct current that can meet an input voltage requirement of the DC/AC converter **202** and store the direct current on two ends of a bus capacitor C on the direct current bus **204** is obtained. The DC/AC converter **202** may convert the direct current on two ends of the bus capacitor C into an alternating current of a corresponding power for use by a load. In this case, the controller **203** may be connected to the DC/DC converter **201** and the DC/AC converter **202**, and is configured to control, based on a result of comparison between the load power (that is, an output power of the DC/AC converter **202**) and the maximum power output by the photovoltaic module **10** in the current illumination environment, the DC/DC converter **201** to switch between different modes.

[0052] It may be understood that, in the photovoltaic system **1** shown in FIG. 4, U.sub.pv represents the output voltage of the photovoltaic module **10**, U.sub.bus represents the bus voltage on the direct current bus **204**, and U.sub.ac represents an output voltage of the DC/AC converter **202**.

[0053] It may be understood that, in the photovoltaic system provided in this disclosure, there may be a plurality of photovoltaic modules connected to the input end of the photovoltaic inverter. The plurality of photovoltaic modules may be connected in series and/or in parallel and then connected to the photovoltaic inverter. FIG. 2 to FIG. 4 are merely examples for description. A quantity of photovoltaic modules in the photovoltaic system is not limited in this disclosure. Optionally, output currents of the plurality of photovoltaic modules may be converged by using a combiner box (that is, the plurality of photovoltaic modules are connected in parallel and then connected to the combiner box) and then provide a direct current input voltage for the photovoltaic inverter. To be

specific, the photovoltaic system may further include the combiner box. The plurality of photovoltaic modules may be connected to the photovoltaic inverter through the combiner box, and the photovoltaic inverter is connected to the alternating current power grid.

[0054] For ease of description, the following uses the structure of the photovoltaic system shown in FIG. 2 to describe function implementation of the photovoltaic inverter and the power adjustment method for the photovoltaic inverter provided in embodiments of this disclosure by using examples.

[0055] In some feasible implementations, the control unit **203** of the photovoltaic inverter **20** may be configured to: when the load power (that is, the output power of the photovoltaic inverter **20**, which is not described below again) is less than the maximum power output by the photovoltaic module **10** in the current illumination environment, control the direct current conversion unit **201** to switch from an MPPT mode to a bus constant mode, so that an input voltage of the direct current conversion unit **201** changes with the load power (that is, the output power of the photovoltaic inverter **20**), and an output power of the photovoltaic module **10** and an output power needed by the photovoltaic inverter **20** keep being adjusted synchronously. The bus constant mode is to adjust an output voltage of the direct current conversion unit **201** to a constant value or to be within a preset range. A lower limit value of any value within the preset range is a peak value of the output voltage of the alternating current inverter unit **202**, and an upper limit value of any value within the preset range may be obtained based on experimental data in an actual application scenario. In other words, the preset range is a range from the lower limit value to the upper limit value.

[0056] It should be noted that when the direct current conversion unit **201** is in the MPPT mode, the direct current conversion unit **201** performs MPPT to maximize an output power of the direct current conversion unit **201**. When performing MPPT, the direct current conversion unit **201** may perturb a port voltage of the photovoltaic module **10** (that is, the input voltage of the direct current conversion unit **201**) at a fixed step size, to perform maximum power tracking. When the direct current conversion unit **201** is in the bus constant mode, the direct current conversion unit **201** adjusts the output voltage of the direct current conversion unit **201** to the constant value or to be within the preset range, so that the bus voltage $U_{sub.bus}$ remains in a constant voltage state. The constant voltage state herein does not mean that the voltage is constant at a value. The constant voltage state herein is relative. Certainly, the constant voltage state may allow the voltage to fluctuate within a specific range.

[0057] In other words, when controlling the direct current conversion unit **201** to switch from the MPPT mode to the bus constant mode, the control unit **203** of the photovoltaic inverter **20** switches from controlling the input voltage of the direct current conversion unit **201** to controlling, by adjusting a duty cycle of a control signal of the direct current conversion unit **201**, the output voltage of the direct current conversion unit **201** to be adjusted to the constant value or to be within the preset range.

[0058] For ease of understanding, the following describes the function implementation of the photovoltaic inverter by using examples in combination with a curve of a power and a voltage (P-U) output by the photovoltaic module and a bus voltage-time ($U_{sub.bus}$ -t) curve, in different cases for the photovoltaic inverter. FIG. 5 is a diagram of a P-U curve of a photovoltaic module according to an embodiment of this disclosure. In FIG. 5, a horizontal coordinate represents the output voltage $U_{sub.pv}$ of the photovoltaic module, that is, an input voltage of the photovoltaic inverter (or the direct current conversion unit); and a vertical coordinate represents the output power P of the photovoltaic module, that is, an input power of the photovoltaic inverter (or the direct current conversion unit). Herein, $P_{sub.max}$ is a maximum power of the photovoltaic module in a specific illumination environment, and a corresponding point c is a maximum power point. FIG. 6 is a diagram of a bus voltage-time curve according to an embodiment of this disclosure. In FIG. 6, a horizontal coordinate represents time (time, denoted as t), a vertical coordinate represents the bus voltage $U_{sub.bus}$, and $U_{sub.bus_ref}$ represents a bus reference voltage value. In one

example, the lower limit value of the preset range may be a minimum value of $U_{sub.bus}$ from $t_{sub.0}$ to $t_{sub.5}$ in FIG. 6, and the upper limit value of the preset range may be a maximum value of $U_{sub.bus}$ from $t_{sub.0}$ to $t_{sub.5}$ in FIG. 6.

[0059] It may be understood that when $t=0$, the photovoltaic inverter **20** is not started.

[0060] In some feasible implementations, when $0 < t \leq t_{sub.0}$, and the load power ($P_{sub.ac}$) is zero, in a start process of the photovoltaic inverter **20**, the direct current conversion unit **201** is in the bus constant mode, and the photovoltaic module **10** works at a point a on the photovoltaic curve (that is, the P-U curve) shown in FIG. 5. The direct current conversion unit **201** may collect the bus voltage $U_{sub.bus}$ of the direct current bus **204**, and may increase the output voltage of the direct current conversion unit **201** to the preset bus reference voltage value $U_{sub.bus_ref}$ based on an input voltage $U_{sub.a}$ provided by the photovoltaic module **10** (in this case, the input voltage $U_{sub.a}$ provided by the photovoltaic module **10** may be close to an open-circuit voltage of the photovoltaic module **10** in the current illumination environment), to implement the bus constant mode. In other words, in the start process of the photovoltaic inverter **20**, even if the load power $P_{sub.ac}$ is zero, the output power P of the photovoltaic module **10** (or the input power of the direct current conversion unit **201**) is not zero, to start the photovoltaic inverter **20**. It may be understood that, in this embodiment of this disclosure, when the load power $P_{sub.ac}$ is zero, the photovoltaic inverter **20** is started by using a very small input power, so that energy consumption can be reduced.

[0061] The bus reference voltage value $U_{sub.bus_ref}$ is a value within the preset range. It may be learned from the foregoing connection relationships shown in FIG. 2 to FIG. 4 that the output voltage of the direct current conversion unit **201** is equal to the bus voltage $U_{sub.bus}$. Therefore, increasing the output voltage of the direct current conversion unit **201** to the bus reference voltage value $U_{sub.bus_ref}$ is equivalent to increasing the bus voltage $U_{sub.bus}$ to the bus reference voltage value $U_{sub.bus_ref}$.

[0062] It may be understood that, when $t_{sub.0} < t < t_{sub.1}$, the load power $P_{sub.ac}$ is still 0, and the bus voltage $U_{sub.bus}$ remains at the bus reference voltage value $U_{sub.bus_ref}$.

[0063] In some feasible implementations, when $t=t_{sub.1}$, the photovoltaic inverter **20** starts to work, and the load power $P_{sub.ac}$ starts to rise. In a working process (after the moment $t_{sub.1}$) of the photovoltaic inverter **20**, when the direct current conversion unit **201** is in the bus constant mode, the direct current conversion unit **201** (which may be specifically the DC/DC control circuit **2012** shown in FIG. 3) may monitor the bus voltage $U_{sub.bus}$ of the direct current bus **204** and the output power of the photovoltaic module **10** in real time. It may be understood that, when the load power $P_{sub.ac}$ changes, the bus voltage $U_{sub.bus}$ fluctuates temporarily. Consequently, the bus voltage $U_{sub.bus}$ is not equal to the preset bus reference voltage value $U_{sub.bus_ref}$. For example, when the load power $P_{sub.ac}$ rises, a phenomenon in which the bus voltage $U_{sub.bus}$ drops transiently occurs. This is because the sudden rise of the load power $P_{sub.ac}$ causes reduction of energy stored on the two ends of the direct current bus **204**, and therefore further causes the phenomenon in which the bus voltage $U_{sub.bus}$ drops transiently. For another example, a drop of the load power $P_{sub.ac}$ causes an increase of energy accumulated on the two ends of the direct current bus **204**, and therefore further causes a phenomenon in which the bus voltage $U_{sub.bus}$ rises transiently.

[0064] When $t_{sub.1} < t \leq t_{sub.2}$, the load power $P_{sub.ac}$ rises to be greater than zero and less than the maximum power $P_{sub.max}$ output by the photovoltaic module **10** in the current illumination environment, and the direct current conversion unit **201** is still in the bus constant mode. When the direct current conversion unit **201** detects that the output voltage of the direct current conversion unit **201** drops to be less than the preset bus reference voltage value $U_{sub.bus_ref}$, the direct current conversion unit **201** may increase the output power of the photovoltaic module **10**, so that the output voltage of the direct current conversion unit **201** rises to the bus reference voltage value $U_{sub.bus_ref}$, to implement the bus constant mode. For example, the direct current conversion unit

201 controls a working point of the photovoltaic module **10** to migrate from the point a to a point b on the photovoltaic curve (that is, the P-U curve) shown in FIG. 5. It may be understood that because the output end of the direct current conversion unit **201** is connected to the direct current bus, the output voltage of the direct current conversion unit **201** is equal to the bus voltage $U_{sub.bus}$. Therefore, increasing the output voltage of the direct current conversion unit **201** to the bus reference voltage value $U_{sub.bus_ref}$ is increasing the bus voltage $U_{sub.bus}$ to the bus reference voltage value $U_{sub.bus_ref}$.

[0065] When $t_{sub.2} < t < t_{sub.3}$, the load power $P_{sub.ac}$ remains unchanged, and the bus voltage $U_{sub.bus}$ remains at the bus reference voltage value $U_{sub.bus_ref}$.

[0066] When $t = t_{sub.3}$, the load power $P_{sub.ac}$ continues to rise, the phenomenon in which the bus voltage $U_{sub.bus}$ drops transiently still occurs.

[0067] When $t_{sub.3} < t < t_{sub.4}$, the load power $P_{sub.ac}$ rises to be greater than or equal to the maximum power $P_{sub.max}$ output by the photovoltaic module **10** in the current illumination environment. In this period (that is, when $t_{sub.3} < t \leq t_{sub.4}$), before the load power $P_{sub.ac}$ rises to be equal to the maximum power $P_{sub.max}$ output by the photovoltaic module **10** in the current illumination environment, the direct current conversion unit **201** is still in the bus constant mode. In addition, when the direct current conversion unit **201** detects that the output voltage of the direct current conversion unit **201** drops to be less than the bus reference voltage value $U_{sub.bus_ref}$, the direct current conversion unit **201** increases the output power of the photovoltaic module **10**. For example, the direct current conversion unit **201** controls the working point of the photovoltaic module **10** to migrate from the point b to the point c on the photovoltaic curve (that is, the P-U curve) shown in FIG. 5. However, when the control unit **203** finds that the load power $P_{sub.ac}$ is greater than or equal to the maximum power $P_{sub.max}$ output by the photovoltaic module **10** in the current illumination environment, the control unit **203** is further configured to control the direct current conversion unit **201** to switch from the bus constant mode to the MPPT mode to perform MPPT on the photovoltaic module **10**. In this case, the working point of the photovoltaic module **10** fluctuates near the point c on the photovoltaic curve (that is, the P-U curve) shown in FIG. 5, that is, fluctuates near the maximum power point.

[0068] When $t = t_{sub.4}$, the load power $P_{sub.ac}$ starts to drop. In a process in which the load power P_{ac} drops, the bus voltage $U_{sub.bus}$ may rise transiently.

[0069] Alternatively, when the direct current conversion unit **201** is in the MPPT mode, and the alternating current inverter unit **202** detects that the output voltage of the direct current conversion unit **201** is not equal to the preset bus reference voltage value $U_{sub.bus_ref}$, the alternating current inverter unit **202** lowers the output power of the alternating current inverter unit **202** to adjust the output voltage of the direct current conversion unit **201** to the bus reference voltage value $U_{sub.bus_ref}$. In this case, the load power $P_{sub.ac}$ may drop because the alternating current inverter unit **202** actively lowers the output power of the alternating current inverter unit **202** when the alternating current inverter unit **202** detects that the output voltage of the direct current conversion unit **201** is not equal to the bus reference voltage value $U_{sub.bus_ref}$ and the direct current conversion unit **201** is in the MPPT mode; or certainly because an input power needed by the load drops. This is not limited in this embodiment of this disclosure.

[0070] It may be understood that, because the output power of the alternating current inverter unit **202** is equal to the load power $P_{sub.ac}$, in the period of $t_{sub.4} < t \leq t_{sub.5}$, when the output power of the alternating current inverter unit **202** is lowered to be less than the maximum power $P_{sub.max}$ output by the photovoltaic module **10** in the current illumination environment, the control unit **203** may control the direct current conversion unit **201** to switch from the MPPT mode back to the bus constant mode. It may be further understood that, after the direct current conversion unit **201** switches back to the bus constant mode, when the direct current conversion unit **201** detects that the output voltage of the direct current conversion unit **201** rises to be greater than the bus reference voltage value $U_{sub.bus_ref}$, the direct current conversion unit **201** may lower the output

power of the photovoltaic module **10**, to lower the output voltage of the direct current conversion unit **201** to the bus reference voltage value $U_{sub.bus_ref}$. For example, the direct current conversion unit **201** controls the working point of the photovoltaic module **10** to migrate from the point c to the point b or from the point c to a point b' on the photovoltaic curve (that is, the P-U curve) shown in FIG. 5.

[0071] Alternatively, when lowering the output power of the photovoltaic module **10**, the direct current conversion unit **201** may control the port voltage of the photovoltaic module **10** to drop to a target voltage value $U_{sub.b}$. The target voltage value $U_{sub.b'}$ is less than a voltage value $U_{sub.c}$ corresponding to the maximum power output by the photovoltaic module **10** in the current illumination environment. In other words, the direct current conversion unit **201** controls the working point of the photovoltaic module **10** to migrate from the point c to the point b' on the photovoltaic curve (that is, the P-U curve) shown in FIG. 5, to lower the output power of the photovoltaic module **10**. In this way, a possibility that the direct current conversion unit **201** and the alternating current inverter unit **202** fail can be reduced. It may be understood that when the load power $P_{sub.ac}$ drops to zero, the direct current conversion unit **201** controls the working point of the photovoltaic module **10** to migrate from the point c to the point a on the photovoltaic curve (that is, the P-U curve) shown in FIG. 5.

[0072] In some feasible implementations, when the direct current conversion unit **201** is in the bus constant mode, and the load power $P_{sub.ac}$ drops and the direct current conversion unit **201** detects that the output voltage of the direct current conversion unit **201** rises to be greater than the bus reference voltage value $U_{sub.bus_ref}$, the direct current conversion unit **201** may lower the output power of the photovoltaic module **10** to lower the output voltage of the direct current conversion unit **201** to the bus reference voltage value $U_{sub.bus_ref}$, to implement the bus constant mode.

[0073] It may be understood that, the photovoltaic module **10** is affected due to reasons such as illumination and shading, and the output power of the photovoltaic module **10** (that is, the input power of the direct current conversion unit **201**) changes with a change of an illumination environment. Therefore, the direct current conversion unit **201** may repeatedly switch between the bus constant mode and the MPPT mode. In this embodiment of this disclosure, there is different implementation logic for the direct current conversion unit **201** in different modes. For details, refer to the foregoing descriptions. Details are not described herein again.

[0074] In this embodiment of this disclosure, the direct current conversion unit **201** and the alternating current inverter unit **202** of the photovoltaic inverter **20** perform power coupling through the direct current bus **204**. When the load power is less than the maximum power output by the photovoltaic module **10** in the current illumination environment, the control unit **203** controls the direct current conversion unit **201** to switch from the MPPT mode to the bus constant mode, so that the input voltage of the direct current conversion unit **201** changes with the load power, and the output power of the photovoltaic module **10** and the output power needed by the photovoltaic inverter **20** keep being adjusted synchronously. In other words, the output power of the direct current conversion unit **201** changes with the output power of the photovoltaic inverter **20**, to improve a power response speed of the direct current conversion unit **201**, so that the output power of the direct current conversion unit **201** can quickly match the output power of the photovoltaic inverter **20**. The bus constant mode is to adjust the output voltage of the direct current conversion unit **201** to the constant value or to be within the preset range.

[0075] FIG. 7 is a schematic flowchart of a power adjustment method for a photovoltaic inverter according to an embodiment of this disclosure. The method is applied to the photovoltaic inverter. The photovoltaic inverter may be any one of the photovoltaic inverters shown in FIG. 2 to FIG. 4. The photovoltaic inverter may include a direct current conversion unit, an alternating current inverter unit, and a direct current bus. An input end of the direct current conversion unit is configured to connect to a photovoltaic module. An output end of the direct current conversion unit is connected to an input end of the alternating current inverter unit through the direct current bus.

An output end of the alternating current inverter unit is configured to connect to a load.

[0076] As shown in FIG. 7, the power adjustment method for the photovoltaic inverter provided in this embodiment of this disclosure may include the following steps.

[0077] **S701:** Detect a load power and a maximum power output by the photovoltaic module in a current illumination environment.

[0078] **S702:** When the load power is less than the maximum power output by the photovoltaic module in the current illumination environment, control the direct current conversion unit to switch from an MPPT mode to a bus constant mode, so that an input voltage of the direct current conversion unit changes with the load power, and an output power of the photovoltaic module and an output power needed by the photovoltaic inverter keep being adjusted synchronously. The bus constant mode is to adjust an output voltage of the direct current conversion unit to a constant value or to be within a preset range.

[0079] In some feasible implementations, after the photovoltaic inverter is started, the photovoltaic inverter may detect the load power and a maximum power output by the photovoltaic module in a current illumination environment in real time. When the load power is less than the maximum power output by the photovoltaic module in the current illumination environment, the direct current conversion unit is controlled to switch from the MPPT mode to the bus constant mode, so that an input voltage of the direct current conversion unit changes with the load power, and the output power of the photovoltaic module and the output power needed by the photovoltaic inverter keep being adjusted synchronously. The bus constant mode is to adjust the output voltage of the direct current conversion unit to the constant value or to be within the preset range. Any value within the preset range may be greater than a peak value of an output voltage of the alternating current inverter unit. In this way, it can be ensured that the alternating current inverter unit can output a current. Any value within the preset range may also be less than a value. The value may be obtained based on experimental data in an actual application scenario.

[0080] It should be noted that when the direct current conversion unit is in the MPPT mode, the photovoltaic inverter performs MPPT to maximize an output power of the photovoltaic inverter. When performing MPPT, the photovoltaic inverter may perturb a port voltage of the photovoltaic module (that is, the input voltage of the direct current conversion unit) at a fixed step size, to perform maximum power tracking. When the direct current conversion unit is in the bus constant mode, the photovoltaic inverter adjusts the output voltage of the direct current conversion unit to the constant value or to be within the preset range, so that a bus voltage of the direct current bus remains in a constant voltage state. The constant voltage state herein does not mean that the voltage is constant at a value. The constant voltage state herein is relative. Certainly, the constant voltage state may allow the voltage to fluctuate within a specific range.

[0081] Therefore, controlling the direct current conversion unit to switch from the MPPT mode to the bus constant mode includes: when controlling the direct current conversion unit to switch from the MPPT mode to the bus constant mode, adjusting a duty cycle of a control signal of the direct current conversion unit, to switch from controlling the input voltage of the direct current conversion unit to controlling the output voltage of the direct current conversion unit to be adjusted to the constant value or to be within the preset range.

[0082] It may be understood that when the load power changes, the bus voltage fluctuates temporarily. For example, when the load power rises, a phenomenon in which the bus voltage drops transiently occurs. Alternatively, when the load power drops, a phenomenon in which the bus voltage rises transiently occurs.

[0083] In some feasible implementations, in a start process of the photovoltaic inverter, when the load power is zero, the direct current conversion unit is in the bus constant mode. The photovoltaic inverter may collect the bus voltage of the direct current bus, and may increase the output voltage of the direct current conversion unit to a preset bus reference voltage value based on an input voltage provided by the photovoltaic module (in this case, the input voltage provided by the

photovoltaic module may be close to an open-circuit voltage of the photovoltaic module in the current illumination environment), to implement the bus constant mode. The bus reference voltage value is a value within the preset range.

[0084] Then, the load power rises. When the load power rises to be greater than zero and less than the maximum power output by the photovoltaic module in the current illumination environment, the direct current conversion unit is still in the bus constant mode. When detecting that the output voltage of the direct current conversion unit drops to be less than a bus reference voltage value, the photovoltaic inverter increases the output power of the photovoltaic module, so that the output voltage of the direct current conversion unit rises to the bus reference voltage value, to implement the bus constant mode.

[0085] Alternatively, when the photovoltaic inverter finds that the load power rises to be greater than or equal to the maximum power output by the photovoltaic module in the current illumination environment, the direct current conversion unit is controlled to switch from the bus constant mode to the MPPT mode to perform MPPT on the photovoltaic module.

[0086] In some feasible implementations, when the direct current conversion unit is in the bus constant mode and the load power drops, and the photovoltaic inverter detects that the output voltage of the direct current conversion unit rises to be greater than the bus reference voltage value, the photovoltaic inverter may lower the output power of the photovoltaic module to lower the output voltage of the direct current conversion unit to the bus reference voltage value, to implement the bus constant mode.

[0087] In some feasible implementations, when the load power is greater than or equal to the maximum power output by the photovoltaic module in the current illumination environment, the direct current conversion unit is controlled to switch from the bus constant mode to the MPPT mode to perform MPPT on the photovoltaic module.

[0088] Alternatively, when the direct current conversion unit is in the MPPT mode, and the photovoltaic inverter detects that the output voltage of the direct current conversion unit is not equal to the bus reference voltage value, the photovoltaic inverter lowers an output power of the alternating current inverter unit, to adjust the output voltage of the direct current conversion unit to the bus reference voltage value. It may be understood that, because the output power of the alternating current inverter unit is equal to the load power, when the output power of the alternating current inverter unit is lowered to be less than the maximum power output by the photovoltaic module in the current illumination environment, the photovoltaic inverter may control the direct current conversion unit to switch from the MPPT mode back to the bus constant mode. It may be further understood that, when the direct current conversion unit is in the bus constant mode, and the photovoltaic inverter detects that the output voltage of the direct current conversion unit rises to be greater than the bus reference voltage value, the photovoltaic inverter may lower the output power of the photovoltaic module to lower the output voltage of the direct current conversion unit to the bus reference voltage value.

[0089] In other words, after the output power of the alternating current inverter unit is lowered to adjust the output voltage of the direct current conversion unit to the bus reference voltage value, the power adjustment method further includes: controlling a port voltage of the photovoltaic module to lower to a target voltage value, to lower the output power of the photovoltaic module. The target voltage value is less than a voltage value corresponding to the maximum power output by the photovoltaic module in the current illumination environment. In this way, a possibility that the direct current conversion unit and the alternating current inverter unit fail can be reduced.

[0090] In this embodiment of this disclosure, when the load power is less than the maximum power output by the photovoltaic module in the current illumination environment, the direct current conversion unit is controlled to switch from the MPPT mode to the bus constant mode, so that the input voltage of the direct current conversion unit changes with the load power, and the output power of the photovoltaic module and the output power needed by the photovoltaic inverter keep

being adjusted synchronously. In other words, the output power of the direct current conversion unit changes with the output power of the alternating current inverter unit, to improve a power response speed of the direct current conversion unit, so that the output power of the direct current conversion unit can quickly match the output power of the alternating current inverter unit.

[0091] The foregoing descriptions are merely specific implementations of this disclosure, but are not intended to limit the protection scope of this disclosure. Any variation or replacement readily figured out by a person skilled in the art within the technical scope disclosed in this disclosure shall fall within the protection scope of this disclosure. Therefore, the protection scope of this disclosure shall be subject to the protection scope of the claims.

Claims

1. A photovoltaic inverter comprising: alternating current (AC) inverter comprising: an AC input end, and an AC output end configured to connect to a load; a direct current (DC) bus; a DC converter comprising: a DC input end configured to connect to a photovoltaic module, and a DC output end connected to the AC input end through the DC bus; and a controller configured to: control, when a load power of the load is less than a maximum power output by the photovoltaic module in a current illumination environment, the DC converter to switch from a maximum power point tracking (MPPT) mode to a bus constant mode such that a DC input voltage of the DC converter changes with the load power, wherein the bus constant mode adjusts a DC output voltage of the DC converter to a constant value or to be within a preset range; maintain synchronous adjustment of a first output power of the photovoltaic module and a second output power needed by the photovoltaic inverter; and control, when the load power is greater than or equal to the maximum power, the DC converter to switch from the bus constant mode to the MPPT in order to perform MPPT on the photovoltaic module.
2. The photovoltaic inverter according to claim 1, wherein the DC converter is further configured to when controlling the DC converter to switch from the MPPT mode to the bus constant mode, switch from controlling the DC input voltage to controlling, by adjusting a duty cycle of a control signal of the DC converter, the DC output voltage to be adjusted to the constant value or to be within the preset range.
3. The photovoltaic inverter according to claim 1, wherein the DC controller is further configured to, when the load power is zero, increase the DC output voltage to a bus reference voltage value based on an input voltage provided by the photovoltaic module in order to implement the bus constant mode, and wherein the bus reference voltage value is greater than a lower limit value of the preset range and less than an upper limit value of the preset range.
4. The photovoltaic inverter according to claim 1, wherein the DC converter is further configured to, when the load power rises to be greater than zero and less than the maximum power and when the DC output voltage drops to be less than a bus reference voltage value, increase the first output power such that the DC output voltage rises to the bus reference voltage value to order to implement the bus constant mode, and wherein the bus reference voltage value is greater than a lower limit value of the preset range and less than an upper limit value of the preset range.
5. The photovoltaic inverter according to claim 4, wherein the DC converter is further configured to, when the load power drops and when the DC output voltage rises to be greater than the bus reference voltage value, lower the first output power such that the DC output voltage drops to the bus reference voltage value in order to implement the bus constant mode.
6. (canceled)
7. The photovoltaic inverter according to claim 1, wherein the AC inverter is configured to, when the DC output voltage is not equal to bus reference voltage value, lower power of the AC inverter in order to adjust the DC output voltage to the bus reference voltage value, and wherein the bus reference voltage value is greater than a lower limit value of the preset range and less than an upper

limit value of the preset range.

8. A method applied to a photovoltaic inverter, wherein the method comprises: detecting a load power of a load and a maximum power output by photovoltaic module in a current illumination environment; controlling, when the load power is less than the maximum power, a direct current (DC) converter to switch from a maximum power point tracking (MPPT) mode to a bus constant mode such that a DC input voltage of the DC converter changes with the load power, wherein the bus constant mode adjusts a DC output voltage of the DC converter to a constant value or to be within a preset range; maintaining synchronous adjustment of a first output power of the photovoltaic module and a second output power needed by the photovoltaic inverter keep being adjusted synchronously; and controlling, when the load power is greater than or equal to the maximum power output by the photovoltaic module in the current illumination environment, the DC converter to switch from the bus constant mode to the MPPT mode in order to perform MPPT on the photovoltaic module.

9. The method according to claim 8, wherein controlling the DC converter to switch from the MPPT mode to the bus constant mode comprises adjusting, when controlling the DC converter to switch from the MPPT mode to the bus constant mode, a duty cycle of a control signal of the DC converter, to switch from controlling the DC input voltage to controlling the DC output voltage to be adjusted to the constant value or to be within the preset range.

10. The method according to claim 8, wherein after controlling the DC converter to switch from the MPPT mode to the bus constant mode, the method further comprises increasing when the load power is zero, the DC output voltage to a bus reference voltage value based on an input voltage provided by the photovoltaic module in order to implement the bus constant mode, and wherein the bus reference voltage value is greater than a lower limit value of the preset range and less than an upper limit value of the preset range.

11. The method according to claim 8, wherein after controlling the DC converter to switch from the MPPT mode to the bus constant mode, the method further comprises when the load power rises to be greater than zero and less than the maximum power and when the DC output voltage drops to be less than a bus reference voltage value, increasing the first output power such that the DC output voltage rises to the bus reference voltage value in order to implement the bus constant mode, and wherein the bus reference voltage value is greater than a lower limit value of the preset range and less than an upper limit value of the preset range.

12. The method according to claim 11, wherein after controlling the DC converter to switch from the MPPT mode to the bus constant mode, the method further comprises when the load power drops and when the DC output voltage rises to be greater than the bus reference voltage value, lowering the first output power such that the DC output voltage drops to the bus reference voltage value in order to implement the bus constant mode.

13. (canceled)

14. The method according to claim 8, wherein after controlling the DC converter to switch from the bus constant mode to the MPPT mode, the method further comprises, when the DC output voltage is not equal to a bus reference voltage value, lowering an output power of an alternating current (AC) inverter in order to adjust the DC output voltage to the bus reference voltage value, and wherein the bus reference voltage value is greater than a lower limit value of the preset range and less than an upper limit value of the preset range.

15. The method according to claim 14, wherein after lowering the output power of the AC inverter in order to adjust the DC output voltage to the bus reference voltage value, the method further comprises controlling a port voltage of the photovoltaic module to drop to a target voltage value, and wherein the target voltage value is less than a voltage value corresponding to the maximum power.

16. A photovoltaic system comprising: a photovoltaic sub-system; and a photovoltaic inverter comprising: an alternating current (AC) inverter comprising: an AC input end; and an AC output

end connect to a load; a direct current (DC) bus; a DC converter comprising: a data DC input end configured to connect to the photovoltaic sub-system; and a DC output end connected to the AC input end through the DC bus; and a controller configured to: control, when a load power of the load is less than a maximum power output by the photovoltaic sub-system in a current illumination environment, the DC converter to switch from a maximum power point tracking (MPPT) mode to a bus constant mode such that a DC input voltage of the DC converter changes with the load power, wherein the bus constant mode adjusts a DC output voltage of the DC converter to a constant value or to be within a preset range; maintain synchronous adjustment first output power of the photovoltaic sub-system and a second output power needed by the photovoltaic inverter; and control, when the load power is greater than or equal to the maximum power, the DC converter to switch from the bus constant mode to the MPPT mode in order to perform MPPT on the photovoltaic sub-system.

17. The photovoltaic system according to claim 16, wherein the controller is further configured to, when controlling the DC converter to switch from the MPPT mode to the bus constant mode, switch from controlling the DC input voltage to controlling, by adjusting a duty cycle of a control signal of the DC converter, the DC output voltage to be adjusted to the constant value or to be within the preset range.

18. The photovoltaic system according to claim 16, wherein the DC converter is configured to, when the load power is zero, increase the DC output voltage to a bus reference voltage value based on an input voltage provided by the photovoltaic sub-system in order to implement the bus constant mode, and wherein the bus reference voltage value is greater than a lower limit value of the preset range and less than an upper limit value of the preset range.

19. The photovoltaic system according to claim 16, wherein the DC converter is configured to, when the load power rises to be greater than zero and less than the maximum power and when the DC output voltage drops to be less than a bus reference voltage value, increase the first output power such that the DC output voltage rises to the bus reference voltage value in order to implement the bus constant mode, and wherein the bus reference voltage value is greater than a lower limit value of the preset range and less than an upper limit value of the preset range.

20. The photovoltaic system according to claim 19, wherein the DC converter is further configured to, when the load power drops and when the DC output voltage rises to be greater than the bus reference voltage value, lower the output power of the photovoltaic sub-system such that the DC output voltage drops to the bus reference voltage value in order to implement the bus constant mode.

21. The photovoltaic system according to claim 16, wherein the AC inverter is configured to, when the DC output voltage is not equal to a bus reference voltage value, lower a third output power of the AC inverter in order to adjust the DC output voltage to the bus reference voltage value, and wherein the bus reference voltage value is greater than a lower limit value of the preset range and less than an upper limit value of the preset range.

22. The photovoltaic system according to claim 21, wherein the controller is further configured to control a port voltage of the photovoltaic sub-system to drop to a target voltage value, and wherein the target voltage value is less than a voltage value corresponding to the maximum power.
